

Exact sensitivity certificates for planar transport maps

Coefficient uncertainty, a convergent box refinement, and a sharp three-atom calibration

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Abstract

An optimal transport map can change substantially even when its target measure moves only a little. We give an exact-arithmetic certificate procedure for finite affine-max transport maps on rational convex polygons. It computes the map discrepancy, separately checks a primal–dual witness for target transport cost, and bounds map discrepancy uniformly over an affine-coefficient box by a possible-winner envelope that needs neither fixed cell combinatorics nor fixed target masses. The envelope relaxes the problem in two distinct ways, and an admissible two-plane family shows that its relative conservatism is unbounded: the envelope is 4 while the sharp maximum is $\varepsilon^2/4$. We prove that the gap is nevertheless linear in the largest coefficient radius, with an explicit constant, and derive from this a convergent box-subdivision refinement that returns a certified interval $[\text{LB}, \text{UB}]$ around the sharp maximum, the lower bound being attained by an explicit realisation. A two-parameter extension of an existing three-atom example gives closed formulas for map error, target distance, and their difference, with a corner maximum on admissible rectangles. A bounded synthetic benchmark compares the envelope with the elementary slope-diameter bound, with explicit realisations, and with the refinement. The contribution is a scoped certificate implementation with proved tightness and convergence statements, not a new general transport-stability exponent.

1 Problem and relationship to prior work

Let $K \subset \mathbb{R}^2$ be a bounded convex polygon with positive area, and let $\rho = |K|^{-1}1_K dx$. Consider potentials

$$u(x) = \max_{1 \leq i \leq m} (a_i \cdot x + c_i), \quad v(x) = \max_{1 \leq j \leq n} (b_j \cdot x + d_j).$$

Within each potential, repeated slopes can be merged by retaining the largest intercept. Write $T_u = \nabla u$, $T_v = \nabla v$ almost everywhere and $\mu = (T_u)_\# \rho$, $\nu = (T_v)_\# \rho$. Our primary quantities are

$$F(u, v) = \|T_u - T_v\|_{L^2(\rho)}^2, \quad W_2(\mu, \nu)^2.$$

They are different: the common-source coupling yields $W_2^2 \leq F$, but it need not be optimal between the targets.

The L^2 distance between optimal maps from a fixed source is the Monge embedding of M  rigot, Delalande and Chazal [8], who relate it to W_2 and, in their supplementary Lemma 5.1, show with two equal-mass atoms rotating on a circle that the map can move like the square root of the target distance even when every atom keeps a fixed positive mass. Delalande and M  rigot [5] prove quantitative stability for a fixed source and a varying target, with exponent 1/6 in the compact convex case; Letrouit [7] shows that for a uniform source on a suitable bounded open set no exponent above 1/3 is possible and conjectures 1/2 for convex sources. Divol, Niles-Weed

Work	What varies	What is controlled	Degeneracies allowed	Quantity certified or bounded
Mérigot et al. [8]; Delalande–Mérigot [5]	target measure	source density, compact support	general targets	$\ T_\mu - T_\nu\ _{L^2}$ by a power of $W_p(\mu, \nu)$
Divol et al. [6]	discrete target, possibly different support	masses, separation, count of atoms	none below the stated lower bounds	$\ T_\mu - T_\nu\ _{L^2} \lesssim W_2^{1/3}$
Bansil–Kitagawa [2]	target masses on fixed sites	site set	zero masses	summed symmetric cell differences by ℓ^1 mass change
An–Lei–Gu [1]	two maps, common source	auxiliary density	–	overlay coupling (may be non-optimal)
Xie et al. [11]	uncertain point sites	distance bounds	–	possible nearest-neighbour sets
This note	affine coefficients in a box	reference potential, polygon	disappearing cells, changing masses, coincident planes	exact F , witnessed W_2^2 , envelope $B \geq \sup F$, interval $[\text{LB}, \text{UB}] \ni \sup F$

Table 1: What varies, what is controlled, and what is certified. The entries for other works are summaries of their stated settings, not reproductions of their theorems.

and Pooladian [6] obtain semi-discrete map stability of order $W_2^{1/3}$ for discrete targets that may have different supports, with constants depending on the masses and the geometry of the atoms. Bansil and Kitagawa [2] give quantitative stability of the cells when masses change on a fixed site set, without a lower bound on individual masses. An, Lei and Gu [1] use the intersections of two source-cell decompositions to construct a target coupling and note that it need not be optimal. Carlier, Delalande and Mérigot [4] study the different problem of a fixed map and a varying source. Reduced-order optimal-value estimators [3] are another distinct certification problem. Possible-winner regions built from upper and lower envelopes of the competing values have a close antecedent in the uncertain Voronoi diagrams of Xie et al. [11], whose partitions record the set of objects that can be nearest to each region; our envelope applies the same screening principle to affine coefficients and integrates a transport cost over it. The scalar function H of Section 4 is a shifted and scaled Huber penalty, the uniform smoothing of the absolute value [10]; its transport role is what is used here.

Our starting calibration comes from OpenAI’s three-affine-max construction [9], which Letrouit’s instability result motivates. We retain its attribution. The calculations below extend that example by allowing both a change of central mass and a tilt, and turn it into a sharp uncertainty test. We do not use or independently validate the source’s claimed general one-third upper bound. The bounded prior-art search is documented in the accompanying package; historical priority for the specialised formulas and envelope is not asserted. Table 1 positions the certificate.

2 Exact map and target certificates

Let C_i and D_j be the maximizing cells of u and v . Ties between distinct slopes lie on finitely many lines, so these cells partition K up to null sets.

Proposition 1 (Optimality and exact overlay). *The map T_u is the unique quadratic optimal map from ρ to μ . The matrix $Q_{ij} = |C_i \cap D_j|/|K|$ has the target masses as its row and column sums, and*

$$F(u, v) = \sum_{i,j} Q_{ij} |a_i - b_j|^2. \quad (1)$$

For rational input coefficients and polygon vertices, all quantities in (1) are rational and computable by polygon clipping.

Proof. For any coupling π of ρ and μ ,

$$\int x \cdot y \, d\pi \leq \int u \, d\rho - \sum_i c_i \mu(\{a_i\}).$$

The graph coupling of T_u attains equality. Maximizing this scalar product minimizes quadratic cost because the marginal second moments are fixed. Equality forces the maximizing slope, unique almost everywhere, which proves uniqueness. Equation (1) follows by integrating its constant integrand on each overlay cell. Rational half-plane intersections have rational vertices, and the shoelace area formula preserves rationality. $\square$

We check target optimality separately. Given a nonnegative matrix P with the prescribed target marginal sums and vectors α, β , require

$$\alpha_i + \beta_j \leq |a_i - b_j|^2, \quad \sum_{i,j} P_{ij} |a_i - b_j|^2 = \sum_i \mu_i \alpha_i + \sum_j \nu_j \beta_j. \quad (2)$$

Weak duality then proves that the displayed cost equals W_2^2 . The checker uses exact equalities and inequalities, rather than an optimizer's status. It rejects a feasible but nonoptimal coupling when the equality fails.

3 A coefficient-box uncertainty certificate

Fix u . For each j , let the two slope coordinates and intercept of the comparison potential vary independently around (b_j, d_j) with radii $(r_{j1}, r_{j2}, r_{j0}) \geq 0$. Define

$$\begin{aligned} e_j(x) &= r_{j1}|x_1| + r_{j2}|x_2| + r_{j0}, \\ U_j(x) &= b_j \cdot x + d_j + e_j(x), \quad L_j(x) = b_j \cdot x + d_j - e_j(x), \\ J(x) &= \{j : U_j(x) \geq L_k(x) \text{ for every } k\}, \\ M_{ij} &= \sum_{\ell=1}^2 (|a_{i\ell} - b_{j\ell}| + r_{j\ell})^2. \end{aligned}$$

Theorem 2 (Uniform map-error enclosure). *For every single global realization $\tilde{v}$ of the coefficient box,*

$$F(u, \tilde{v}) \leq B := \frac{1}{|K|} \sum_i \int_{C_i} \max_{j \in J(x)} M_{ij} \, dx. \quad (3)$$

For rational box data, B is exactly computable by a finite arrangement of lines and rational polygon areas. No preservation of cell combinatorics, positive individual cell masses, or fixed target masses is assumed.

Proof. If j wins at x for a realization, then its realized value is at most $U_j(x)$, while every competing realized value is at least $L_k(x)$. Thus $j \in J(x)$. The coordinatewise triangle inequality bounds the squared slope discrepancy by M_{ij} on C_i , proving the pointwise inequality and then (3).

On any coordinate orthant, each $U_j - L_k$ is affine. Partition each C_i intersected with that orthant by their zero lines. The set $J(x)$ is constant on the interior of each positive-area piece; its contribution is that piece's rational area times a rational constant. Boundaries are null. Identically zero comparisons cause no split. For any fixed realization, coincident affine functions have identical slopes; ties between unequal slopes remain null. These degeneracies do not invalidate the integral. $\square$

Three elementary consequences are used by the implementation: $J(x)$ is never empty, because a nominal maximiser j satisfies $U_j \geq b_j \cdot x + d_j \geq b_k \cdot x + d_k \geq L_k$; with all radii zero, B equals the exact discrepancy $F(u, v)$ almost everywhere; and the arithmetic mean of the vertices of a positive-area convex polygon lies in its interior, so it is a valid sign representative for each arrangement region. We also record the elementary slope-diameter bound

$$B_0 := \sum_i \rho(C_i) \max_j M_{ij} \geq B, \quad (4)$$

which ignores the winner screening; the benchmark of Section 5 measures what the screening buys.

3.1 Two relaxations, and an adverse family

The envelope relaxes the problem in two distinct ways. First, M_{ij} maximises the slope discrepancy of plane j over the whole box without requiring that the same coefficient choice makes j win at x . Second, the pointwise maximum lets different source points use different global parameters. The following family, supplied by the external review of version 0.1.0 and rederived here, shows that the first relaxation costs an unbounded factor while the second costs exactly a further factor two.

Proposition 3 (Adverse family). *Let $K = [1, 2] \times [0, 1]$, $u(x) = x_1$, and let the comparison box consist of the fixed plane x_1 and the plane $bx_1 + \varepsilon$ with nominal $b = 0$, slope radius 1 in the first coordinate and all other radii zero, where $0 \leq \varepsilon \leq 1$ is rational. A realisation is $v_b(x) = \max\{x_1, bx_1 + \varepsilon\}$ with one global $b \in [-1, 1]$. Put $d = 1 - b \in [0, 2]$. Then*

$$F(u, v_b) = \begin{cases} d^2, & 0 \leq d \leq \varepsilon/2, \\ \varepsilon d - d^2, & \varepsilon/2 \leq d \leq \varepsilon, \\ 0, & d \geq \varepsilon, \end{cases} \quad \sup_{b \in [-1, 1]} F(u, v_b) = \frac{\varepsilon^2}{4}, \quad (5)$$

the supremum being attained only at the interior parameter $b = 1 - \varepsilon/2$ when $\varepsilon > 0$, while both endpoints $b = \pm 1$ give $F = 0$. The envelope of Theorem 2 and the baseline (4) both equal 4 for every such ε , with a single arrangement region. The pointwise supremum that respects winner-slope compatibility but still lets b depend on x integrates to $\varepsilon^2/2$. Hence

$$\frac{\varepsilon^2}{4} = \sup_b F(u, v_b) \leq \frac{\varepsilon^2}{2} \leq 4 = B,$$

the first inequality costs exactly a factor 2 in this family and the second a factor $8/\varepsilon^2$, so the envelope-to-maximum ratio $16/\varepsilon^2$ is unbounded with the slope radius fixed, and at $\varepsilon = 0$ every realisation has $F = 0$ while $B = 4$.

Proof. Plane 2 wins where $dx_1 < \varepsilon$. For $d > 0$ its cell is $\{1 \leq x_1 < \min(2, \varepsilon/d)\} \times [0, 1]$, of mass $(\min(2, \varepsilon/d) - 1)_+$ since $|K| = 1$, and its squared slope discrepancy is $(1 - b)^2 = d^2$, which gives (5); at $d = 0$ the plane has the reference slope and $F = 0$ after merging. The branch d^2 increases on $[0, \varepsilon/2]$ and $\varepsilon d - d^2$ decreases on $[\varepsilon/2, \varepsilon]$, so the supremum is $\varepsilon^2/4$ at $d = \varepsilon/2$. For every $x \in K$, $U_2(x) = x_1 + \varepsilon \geq x_1 = L_1(x)$, so $2 \in J(x)$ everywhere; $M_{12} = (1 + 1)^2 = 4$ and $M_{11} = 0$, so the integrand is 4 everywhere, and $B_0 = 4$ likewise. At a point with $x_1 \in (1, 2)$, plane 2 wins exactly for $b > 1 - \varepsilon/x_1$, and the supremum of $(1 - b)^2$ over these b is $(\varepsilon/x_1)^2$; integrating, $\varepsilon^2 \int_1^2 x_1^{-2} dx_1 = \varepsilon^2/2$. $\square$

The family also shows that the corner reduction of Corollary 8 below cannot extend to general coefficient boxes: both endpoints give error zero while an interior parameter gives the maximum. The loss is unrelated to computational scale, since one region suffices. It is a limitation of informativeness, not a counterexample to Theorem 2.

3.2 Tightness at small radius

Let r be the largest radius entry, v the nominal realisation (duplicate slopes merged, largest intercept kept; in the proof the nominal winner at a point is an original index with the winning slope and the largest intercept, so that J , M_{ij} and the pair count use original indices), $R_K = \max_{x \in K}(|x_1| + |x_2| + 1)$, D_K any upper bound for the Euclidean diameter of K (the implementation uses the ℓ^1 vertex diameter), and

$$\Lambda = \max_{i,j} \sum_{\ell=1}^2 (|a_{i\ell} - b_{j\ell}| + r_{j\ell}),$$

an ℓ^1 bound for the distance between any reference slope and any realised comparison slope. Norms of slopes are Euclidean.

Theorem 4 (Tightness).

$$0 \leq B - F(u, v) \leq \gamma(r) := 2r\Lambda + 2r^2 + \frac{4rR_K D_K n(n-1)\Lambda}{|K|}. \quad (6)$$

Proof. The lower inequality is Theorem 2 for the nominal realisation. Fix x in the interior of some C_i , off every nominal tie line, and let k be the nominal winner at x . For $j \in J(x)$ write

$$M_{ij} - |a_i - b_k|^2 = (M_{ij} - |a_i - b_j|^2) + (|a_i - b_j|^2 - |a_i - b_k|^2).$$

The first bracket equals $\sum_{\ell} (2r_{j\ell}|a_{i\ell} - b_{j\ell}| + r_{j\ell}^2) \leq 2r\Lambda + 2r^2$. The second vanishes for $j = k$. For $j \neq k$ in $J(x)$, the membership $U_j(x) \geq L_k(x)$ gives $g_k(x) - g_j(x) \leq e_j(x) + e_k(x) \leq 2rR_K$, where $g_j = b_j \cdot x + d_j$, and $g_k(x) \geq g_j(x)$ because k wins; so x lies in $T_{jk} := \{y \in K : 0 \leq g_k(y) - g_j(y) \leq 2rR_K\}$. Moreover $|a_i - b_j|^2 - |a_i - b_k|^2 = (b_k - b_j) \cdot (2a_i - b_j - b_k) \leq 2\Lambda|b_k - b_j|$, since a Euclidean norm is at most the ℓ^1 norm bounded by Λ . Integrating over K ,

$$|K|(B - F(u, v)) \leq (2r\Lambda + 2r^2)|K| + \sum_{j \neq k} |T_{jk}| 2\Lambda|b_k - b_j|.$$

A pair with $b_j = b_k$ contributes nothing. Otherwise T_{jk} lies in a closed strip of width $2rR_K/|b_k - b_j|$, and a strip of width w meets a convex set of diameter at most D_K in area at most wD_K (integrate the chord lengths across the strip). Each ordered pair therefore contributes at most $4rR_K D_K \Lambda$, and there are at most $n(n-1)$ of them. $\square$

Remark 5 (Margins and slope separation). Let $\mu(x)$ be the nominal margin: the winning value minus the largest competing value among planes with a different slope. Where $\mu(x) > 2rR_K$, $J(x)$ contains only planes with the winner's slope and the excess is at most $2r\Lambda + 2r^2$; elsewhere it is at most $\max_{i,j} M_{ij}$. Hence $B - F(u, v) \leq 2r\Lambda + 2r^2 + (\max_{i,j} M_{ij}) |\{\mu \leq 2rR_K\}|/|K|$, and the strip bound gives $|\{\mu \leq t\}| \leq n(n-1)D_K t/\sigma$ with σ the smallest distance between distinct comparison slopes. The uniform constant in (6) avoids σ because the strip width and the cost difference are reciprocal in $|b_k - b_j|$. With the ℓ^1 vertex diameter as D_K , all quantities in γ are rational for rational data; the test suite checks (6) on 108 random boxes.

3.3 A convergent refinement

Theorem 2 applies to every sub-box of the coefficient box, and Theorem 4 says that the envelope of a sub-box of radius r overshoots the exact error at its centre by $O(r)$. This gives a certificate that retains the global parameter dependence.

Theorem 6 (Convergent box subdivision). *Let the coefficient box be covered by finitely many sub-boxes P of the same form, with largest radius r_* . Put $\text{UB} = \max_P B(P)$ and let LB be the largest value of $F(u, \cdot)$ over any finite set of realisations in the box that contains every sub-box centre $c(P)$. Then*

$$\text{LB} \leq \sup_{\tilde{v}} F(u, \tilde{v}) \leq \text{UB}, \quad \text{UB} - \text{LB} \leq \max_P [B(P) - F(u, v_{c(P)})] \leq \gamma_{\text{box}}(r_*),$$

where γ_{box} is (6) with Λ computed for the full box.

Proof. Every realisation lies in some P , so Theorem 2 for that sub-box gives $F \leq B(P) \leq \text{UB}$; LB is a maximum of feasible values. For the gap, $\text{UB} - \text{LB} \leq B(P^*) - F(u, v_{c(P^*)})$ for a sub-box P^* attaining UB , and Theorem 4 applies to P^* with its own constant, which is at most γ_{box} because for a sub-interval $[b' - r', b' + r']$ of $[b - r, b + r]$ one has $|a - b'| + r' \leq |a - b| + |b - b'| + r' \leq |a - b| + r$. $\square$

The implementation (`refine_bound`) keeps a best-first heap of sub-boxes ordered by their envelopes, splits the sub-box with the largest envelope along its widest coordinate, evaluates the exact F at each child centre and at the child's outer endpoint along the split coordinate (literal points of the box, which supply LB ; the attaining point is returned both as coefficients and as its merged equivalent), and never splits a sub-box whose envelope is at most LB ; pruning preserves both inequalities because such sub-boxes cannot exceed LB . It stops when the largest remaining envelope is at most LB , when $\text{UB} - \text{LB}$ is within a requested tolerance, or when the box budget is exhausted; in every case it returns the valid interval, the number of sub-boxes evaluated and the stop reason. Splitting the widest coordinate halves the largest radius of a lineage at least once every $3n$ splits, so for any positive tolerance only finitely many sub-boxes can remain above it and the loop terminates; exact convergence occurs only when a probe meets a maximiser, which bisection from $[-1, 1]$ never does at the non-dyadic maximiser $b = 19/20$ of the adverse family. Uniform refinement needs a number of sub-boxes exponential in the box dimension $3n$; the benchmark records the consequence. On the adverse family the interval reaches $[\varepsilon^2/4 \cdot (1 - 10^{-5}), \varepsilon^2/4 \cdot (1 + 1.4 \cdot 10^{-3})]$ with 256 sub-boxes, where the envelope was 4.

4 A sharp two-parameter calibration

Let $d \geq 2$, let ρ be uniform on $[-1, 1]^d$, and assume

$$0 < a < 1, \quad 0 < a + s - |b|, \quad a + s + |b| < 1. \quad (7)$$

Define

$$u_a(x) = \max\{-x_1, x_1, a\}, \quad v_{a,s,b}(x) = \max\{-x_1, x_1, a + s + bx_2\}.$$

The central strips have slice half-widths a and $a + s + bt$ at $x_2 = t$. Their target laws are

$$\begin{aligned} \mu_a &= \frac{1-a}{2}(\delta_{-e_1} + \delta_{e_1}) + a\delta_0, \\ \nu_{a,s,b} &= \frac{1-a-s}{2}(\delta_{-e_1} + \delta_{e_1}) + (a+s)\delta_{be_2}. \end{aligned}$$

Theorem 7 (Exact error and coupling excess). *Under (7), set*

$$H(s, b) = \begin{cases} |s|, & |s| \geq |b|, \\ (s^2 + b^2)/(2|b|), & |s| < |b|. \end{cases}$$

Then

$$F(u_a, v_{a,s,b}) = H(s, b) + (a + s)b^2, \quad (8)$$

$$W_2(\mu_a, \nu_{a,s,b})^2 = |s| + (a + s)b^2. \quad (9)$$

In particular the overlap coupling is target-optimal exactly when $|s| \geq |b|$. Its excess cost is

$$F - W_2^2 = \begin{cases} 0, & |s| \geq |b|, \\ (|b| - |s|)^2/(2|b|), & |s| < |b|. \end{cases} \quad (10)$$

Proof. The central intervals are nested on each slice. Their symmetric difference has conditional source probability $|s + bt|$. The first-coordinate squared map difference is the indicator of that symmetric difference; the second-coordinate contribution is b^2 on the new central strip. Consequently $F = \mathbb{E}|s + bU| + (a + s)b^2$, with U uniform on $[-1, 1]$. Integrating the absolute value, splitting at $-s/b$ when it is internal, gives H and (8). (For $|b| > 0$, $H(s, b) = h_{|b|}(s)/|b| + |b|/2$ with h_λ the Huber penalty [10].)

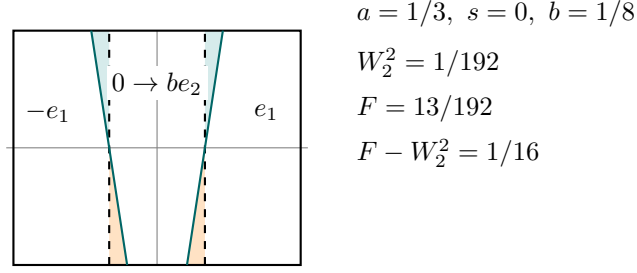
Order the atoms (left, centre, right), so that the cost matrix is $\begin{pmatrix} 0 & 1+b^2 & 4 \\ 1 & b^2 & 1 \\ 4 & 1+b^2 & 0 \end{pmatrix}$. For $s \geq 0$ the coupling keeps the common mass on each atom pair, namely $\frac{1-a-s}{2}$ on left→left, a on centre→centre (the centre moves from 0 to be_2 , cost b^2), $\frac{1-a-s}{2}$ on right→right, and moves $s/2$ from each outer atom to the new centre at cost $1 + b^2$ each. For $s < 0$ it keeps $\frac{1-a}{2}$, $a + s$, $\frac{1-a}{2}$ on the diagonal and moves $-s/2$ from the old centre to each outer atom at cost 1 each. Horizontal cost is $|s|$ and vertical cost is $(a + s)b^2$. Exact feasible duals in (2) are

	α	β
$s \geq 0$	$(1, 0, 1)$	$(-1, b^2, -1)$
$s < 0$	$(-1, 0, -1)$	$(1, b^2, 1)$.

Their objective equals the constructed cost, proving (9). Subtracting gives (10), including $s = b = 0$. Coordinates beyond the first two integrate out. $\square$

Corollary 8 (Sharp parameter-rectangle maximum). *Fix a and a closed rectangle $[s_-, s_+] \times [b_-, b_+]$ satisfying (7) throughout. The maximum of F is the largest of its four corner values. It is enough to check admissibility at the corners.*

Proof. For fixed b , $\mathbb{E}|s + bU|$ is convex in s , and $(a + s)b^2$ is affine. For fixed s , the first term is convex in b and the second is convex because $a + s > 0$. Maximizing in the two coordinates successively reduces the rectangle to its corners. Joint convexity is not required. The admissibility margins $a + s - |b|$ and $1 - a - s - |b|$ take their minima over the rectangle at corners as well. $\square$



Source cells: reassigned mass is shaded

Figure 1: The targets have unchanged atom masses, yet their maps disagree on source strips. Dashed boundaries belong to the reference map; solid boundaries to the tilted map. Labels denote map values, not target points in the source plane. Diagram coordinates are rounded for display; all reported values are exact.

For $s = 0$, $b \in [-\beta, \beta]$, and $0 < \beta < \min(a, 1 - a)$, the exact maximum is $\beta/2 + a\beta^2$. The envelope of Theorem 2, applied with only the central vertical slope uncertain, is $\beta + a\beta^2$: the inner ambiguous source strips have total mass $\beta/2$ and worst squared cost one; the outer ambiguous strips have mass $\beta/2$ and cost $1 + \beta^2$; the common central region has mass $a - \beta/2$ and cost β^2 . The gap $\beta/2$ is the price of the pointwise relaxation in this family; the pointwise maxima are attainable on each region, so the winner-slope relaxation costs nothing here, in contrast with Proposition 3. The refinement of Theorem 6 closes the gap with three sub-boxes, because the endpoint probes meet the corner maximum.

Setting $s = 0$, $b = a/2$ and $0 < a < 1/2$ recovers the source’s degenerating-mass example [9]: $W_2 = a^{3/2}/2$ and $\sqrt{F} = \sqrt{a(1 + a^2)}/2$. Keeping a fixed instead gives $W_2 = \sqrt{a}|b|$ and $\sqrt{F} \sim \sqrt{|b|/2}$. Thus even with fixed positive atom masses, this family has square-root rather than Lipschitz map sensitivity as the central site moves; the phenomenon itself is already in [8] (supplementary Lemma 5.1, two rotating atoms). These statements are direct consequences of the calibration, not new general upper bounds.

5 A bounded benchmark

Table 2 and Table 3 summarise the 65 cases recorded in `evidence/benchmark.json`, computed by the shipped `benchmark.py` on one machine (Python 3.14, Apple M2; process peak memory 23 MB; every exact field is recomputed and compared by its `--check` mode). The families are: the tilt calibration with nine (a, β) pairs, where the exact maximum is known; the adverse family of Proposition 3 for three ε ; and generic random rational potentials with $n \in \{2, 3, 4, 6, 8\}$ comparison planes, n reference planes, all $3n$ coefficients uncertain with a common radius $r \in \{1/64, 1/16, 1/4\}$, on a square, a triangle and a pentagon, plus cases with dyadic denominators 2^8 and 2^{20} , an inactive plane, a near-coincident plane, and a region-cap refusal. The refinement budget is 32 sub-boxes for $n \leq 6$ and 8 for $n = 8$.

Three observations are supported. First, the winner screening is what makes the envelope informative: B/B_0 falls from 0.74–1.00 at $n = 2$ to 0.12–0.46 at $n = 6$ and 0.15–0.42 at $n = 8$. Second, at small radius the envelope is nearly attained: at $r = 1/64$ explicit realisations reach 0.95–1.00 of B , at $r = 1/16$ 0.83–0.99, and at $r = 1/4$ 0.56–0.97, in line with the linear-in- r statement of Theorem 4; the inactive plane, the near-coincident plane and the long denominators change the region count, not the tightness. Third, subdivision is decisive on low-dimensional boxes (the

Family	B/B_0	$B/\sup F$	LB/ $\sup F$	UB/ $\sup F$	boxes	s
tilt family, nine (a, β) pairs	0.01–0.05	1.89–1.98	1.0000	1.0000	3	0.02
adverse family, $\varepsilon = 1/2$, budget 64	1.00	64.00	1.0000	1.0163	63	0.06
adverse family, $\varepsilon = 1/2$, budget 256	1.00	64.00	1.0000	1.0011	255	0.28
adverse family, $\varepsilon = 1/10$, budget 64	1.00	1600	1.0000	1.0269	63	0.06
adverse family, $\varepsilon = 1/10$, budget 256	1.00	1600	1.0000	1.0012	255	0.28
adverse family, $\varepsilon = 1/100$, budget 64	1.00	160000	1.0000	1.0348	63	0.07
adverse family, $\varepsilon = 1/100$, budget 256	1.00	160000	1.0000	1.0014	255	0.31

Table 2: Families with a known sharp maximum. B is the envelope, B_0 the slope-diameter bound (4), [LB, UB] the refinement interval at the stated budget; ranges are over the cases in the row. Wall time is the envelope plus the refinement.

Cases	B/B_0	LB/ B	UB/ B	regions	s
$n = 2, r = 1/64$, three polygons	0.74–1.00	1.00	1.00	14	0.10
$n = 2, r = 1/16$, three polygons	0.76–1.00	0.98–0.99	0.99–1.00	14	0.10
$n = 2, r = 1/4$, three polygons	0.86–1.00	0.90–0.97	0.95–1.00	10	0.09
$n = 3, r = 1/64$, three polygons	0.30–0.46	0.96–0.99	0.99–1.00	20	0.18
$n = 3, r = 1/16$, three polygons	0.32–0.49	0.87–0.94	0.97–0.99	23	0.21
$n = 3, r = 1/4$, three polygons	0.48–0.63	0.68–0.69	0.91–0.98	20	0.19
$n = 4, r = 1/64$, three polygons	0.19–0.70	0.98–0.99	1.00	57	0.63
$n = 4, r = 1/16$, three polygons	0.19–0.73	0.91–0.97	0.98–1.00	57	0.68
$n = 4, r = 1/4$, three polygons	0.21–0.86	0.72–0.86	0.93–1.00	47	0.59
$n = 6, r = 1/64$, three polygons	0.12–0.32	0.96–0.98	1.00	247	3.21
$n = 6, r = 1/16$, three polygons	0.13–0.34	0.86–0.94	0.99–1.00	243	3.40
$n = 6, r = 1/4$, three polygons	0.15–0.46	0.57–0.80	0.96–1.00	218	4.01
$n = 8, r = 1/64$, three polygons	0.15–0.28	0.95–0.98	1.00	679	2.84
$n = 8, r = 1/16$, three polygons	0.17–0.30	0.83–0.93	0.99–1.00	652	2.75
$n = 8, r = 1/4$, three polygons	0.26–0.42	0.56–0.75	0.96–1.00	407	1.88
$n = 4, r = 1/16$, denominators 2^8	0.56	0.93	0.99	71	0.65
$n = 4, r = 1/16$, denominators 2^{20}	0.46	0.93	0.99	101	1.17
$n = 4$ plus an inactive plane, $r = 1/16$	0.66	0.95	0.99	93	1.15
$n = 4$ plus a near-coincident plane, $r = 1/16$	0.66	0.94	0.99	168	1.80
$n = 8, r = 1/4$, region cap 100	–	–	–	refused	–

Table 3: Generic random boxes with all $3n$ coefficients uncertain. LB is the largest exact error among the explicit realisations probed by the refinement; UB its refined envelope at the budget; regions is the largest arrangement size of the envelope; s the largest wall time in the row.

one-parameter tilt and adverse families converge within three and 256 sub-boxes) and marginal at a budget of 32 on $3n$ -dimensional boxes, where halving one coordinate barely changes the envelope; this is the exponential cost predicted after Theorem 6, and the certified lower bound is then the useful output. The largest envelope took 0.52 s with 679 regions; the region cap refuses rather than returning a partial sum. These are single-machine observations, not performance claims.

6 Implementation, checks, and limitations

The accompanying Python implementation uses only the standard library and exact rational arithmetic. Input floats, malformed rational strings such as $1/0$, repeated reference slopes, invalid polygons, invalid marginal sums, infeasible duals, and nonzero primal–dual gaps are rejected with a diagnostic. Repeated slopes are merged explicitly by `merge_planes` before overlay. The uncertainty routines permit coinciding comparison planes and disappearing cells. The envelope remains an upper bound, not necessarily a realizable maximum; the refinement returns a certified interval whose lower end is realised.

Tests compare the clipping implementation with a separately implemented half-plane vertex-enumeration and convex-hull calculation. They cover 45 exact calibration cases, 24 generic polygon/potential pairs, invariances, zero-radius exactness, the sharp-versus-enclosed uncertainty gap, a 169-point rectangle grid, the adverse family on a 101-point grid for four values of ε , the tightness inequality (6) on 108 random boxes, the monotone convergence and feasibility of the refinement, and the rejection of malformed inputs. Thirty-two sampled box realizations provide additional falsification tests, not proof of a uniform bound. Semantic negative controls include a feasible nonoptimal overlap coupling and corrupted primal or dual witnesses. The test suite runs in normal and optimized Python modes; it does not rely on removable assertions. The command-line `--check` mode recomputes every recorded field with the same implementation: it is a reproducibility check, not an independent verification.

The mathematical claims rest on the proofs above, while tests support the implementation. Neither is formal proof-assistant verification or external peer review. Version 0.1.0 of this note, which lacked Propositions 3, Theorems 4–6, Table 1 and the benchmark, received a fresh-context internal review and an external review supplied by the publisher; the response to the latter accompanies the package. The code addresses rational planar geometry and a uniform source, not high-dimensional scaling, arbitrary source densities, or reconstruction of potentials with prescribed masses. The proofs of Theorems 2, 4 and 6 do not use the dimension, but no higher-dimensional or weighted integration backend is implemented, because no concrete application justified it. These are material limits, not presumed solved by the general transport literature.

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